Skills Summary

Reading the Number in the Right Place

Use this reference while completing your worksheet or when studying.

Skill 1 — Across first, then up

Rule: in (x, y) the **first** number counts across from the corner and the **second** counts up.

- Say it as you plot: "along the hall, then up the stairs."
- The same two numbers in the other order give a *different* point, usually far away — not a small slip.
- To count grid steps between two points: add the across difference and the up difference.

Example: How many grid steps apart are (3, 8) and (8, 3), moving only along grid lines?
Step 1. The two pairs use the same numbers in opposite slots, so they are two different points — count the across move and the up move separately, then add them.
Step 2. Across: 3 to 8 is 5 steps. Up: 8 down to 3 is 5 steps. Total $5 + 5 =$ **10** steps.

Try it: How many grid steps apart are (2, 5) and (5, 2), moving only along grid lines?

check: **6**

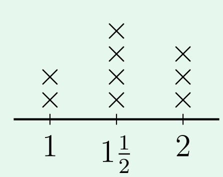
(!) Watch out: "Same two numbers" never means "same point". Plot both and look — they sit on opposite sides of the diagonal.

Skill 2 — Counting marks on a line plot

Two numbers, two jobs. The number *under* the line says what was measured. The marks *above* it say how many.

- "How many?" → count marks. "How long / how tall?" → read the number under the line.
- To find how many were measured altogether, add every stack.

Example: Each × is one ribbon. How many ribbons are $1\frac{1}{2}$ metres long?



Step 1. The question asks how many *ribbons*, so the answer comes from counting marks, not from the number printed under the line.
Step 2. Find $1\frac{1}{2}$ on the line and count the marks stacked above it: there are **4**.

Try it: On another line plot, 2 marks stand above the 5, 5 marks above the 6 and 3 marks above the 7. How many things measured 7?

check: **3**

Skill 3 — Ordered pairs from two patterns

Rule: whichever pattern is named *first* supplies the *first* number of every pair.

- Write both patterns out in a row before pairing them up, and match term 1 with term 1, term 2 with term 2, and so on.
- Getting both numbers right is only half the job — the order carries the meaning.

Example: Pattern P starts at 2 and adds 4. Pattern Q starts at 1 and adds 2. Write the 3rd pair (P, Q).
Step 1. P is named first, so P's third term goes in the first slot and Q's third term in the second — list both patterns before pairing anything.
Step 2. P: 2, 6, 10, so the first number is **10**. Q: 1, 3, 5, so the second number is **5**. The pair is (10, 5).

Try it: Pattern P starts at 5 and adds 5. Pattern Q starts at 1 and adds 3. Write the 4th pair (P, Q).

check: **20** then **10**